

NETFLIX



DATA CASE STUDY

Netflix Content Strategy: Data Exploration & Visualization

An end-to-end EDA of Netflix's global content catalogue — cleaning, univariate and bivariate analysis, and outlier checks — turned into business insights on content strategy across markets.

8,807 TITLES

12 ATTRIBUTES

11 VISUAL ANALYSES

TOOLS USED

Python

NumPy

Pandas

Matplotlib

Seaborn

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A Global Netflix Content Catalogue

8,807 titles across 12 attributes covering content type, cast, director, country, genre, rating, release year, and platform add date — built for univariate, bivariate, and outlier analysis.

Identity

show_id, type, title

People

director, cast

Geography

country

Platform Timing

date_added, release_year

Classification

rating, listed_in (genre)

Format

duration, description

11 Analyses, 3 Stages

From single-variable distributions to cross-variable relationships and outlier checks, the analysis builds up through cleaning, visualization, and business interpretation.

UNIVARIATE · 6

- Content Type: Movies vs TV Shows
- Distribution of Release Year
- Distribution of Movie Duration
- Distribution of TV Show Seasons
- Content Count by Maturity Rating
- Top 10 Countries by Content Count

BIVARIATE · 3

- Movie Duration by Maturity Rating
- TV Show Seasons by Maturity Rating
- Release Year by Content Type

OUTLIER CHECK · 2

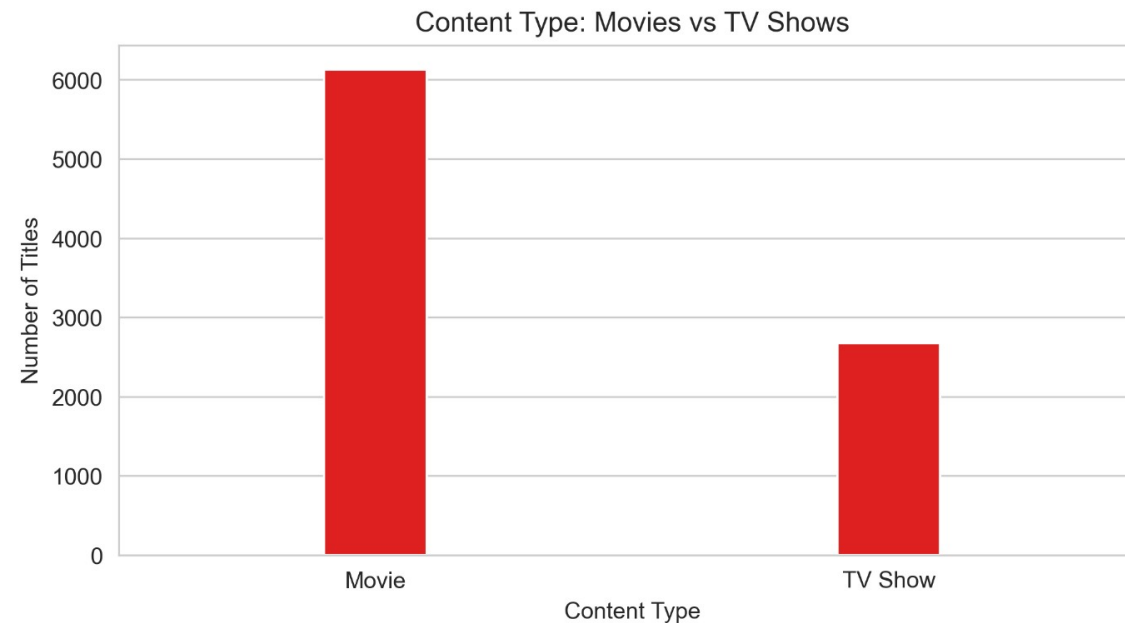
- Outlier Detection: Movie Duration
- Outlier Detection: TV Show Seasons

Content Type: Movies vs TV Shows

A simple count of titles by type establishes the baseline shape of the catalogue.

RESULT Movies dominate the catalogue — 6,131 titles (69.6%) vs 2,676 TV Shows (30.4%).

APPROACH Counted rows by the type column and plotted Movie vs TV Show as a bar chart to see the overall split.

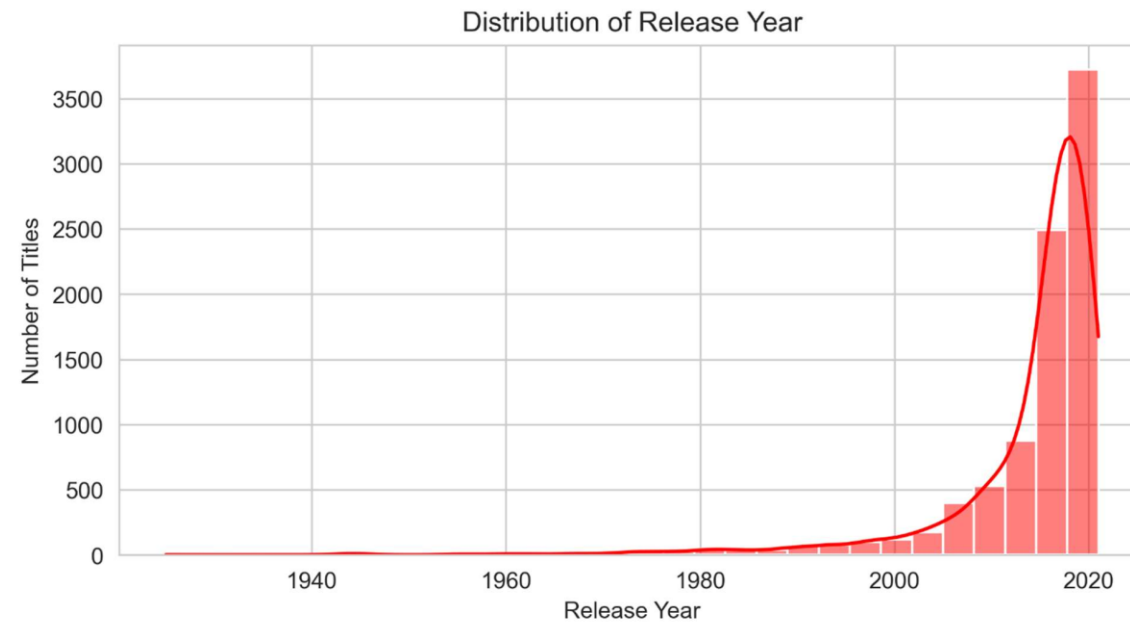


Distribution of Release Year

A histogram of release_year shows how old (or new) the catalogue's content is.

RESULT Strongly right-skewed: very few titles before 2000; the median release year is 2017.

APPROACH Plotted a histogram with a KDE overlay on release_year across the full 1925–2021 range.

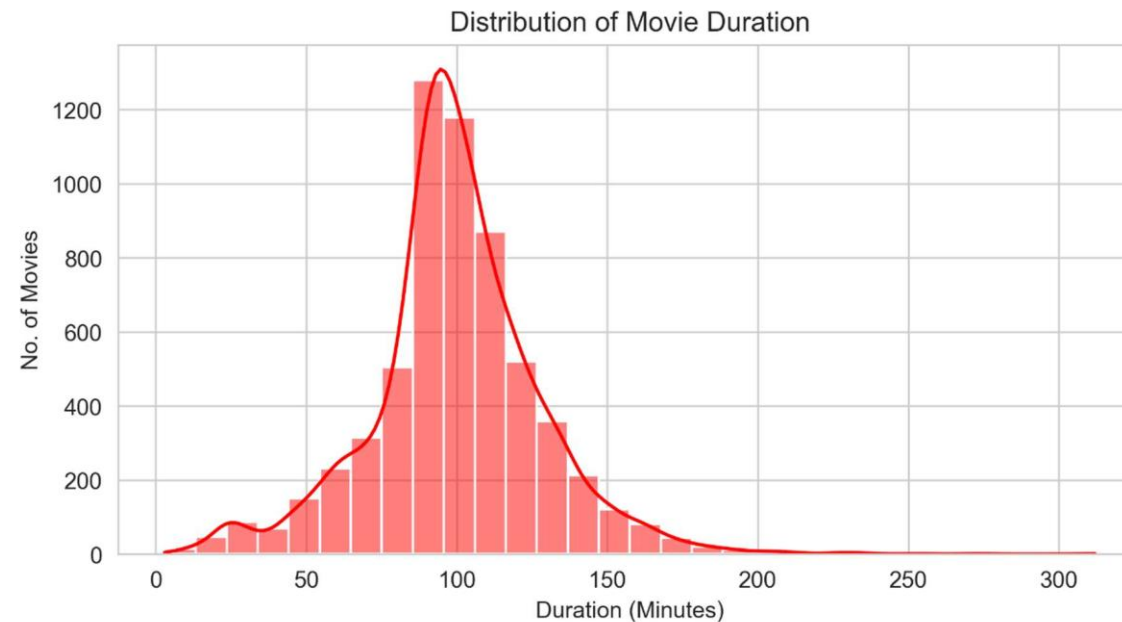


Distribution of Movie Duration

Movie runtimes were parsed from the duration column and plotted as a histogram.

RESULT Most movies run 87–114 minutes, centered around ~98 minutes — a standard feature length.

APPROACH Extracted the numeric minute value from duration for rows where type = Movie, then plotted the distribution.

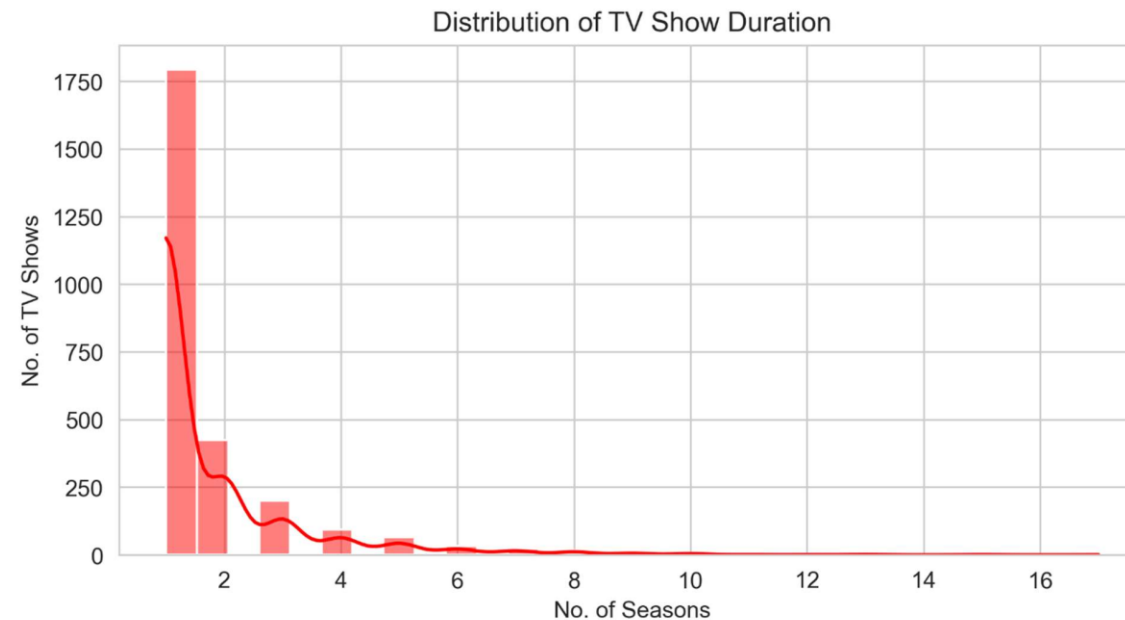


Distribution of TV Show Seasons

Season counts were parsed from duration for TV Shows to see renewal patterns.

RESULT About 67% of TV Shows have just 1 season; the longest-running show has 17 seasons.

APPROACH Extracted the season count from duration where type = TV Show and plotted it as a histogram.

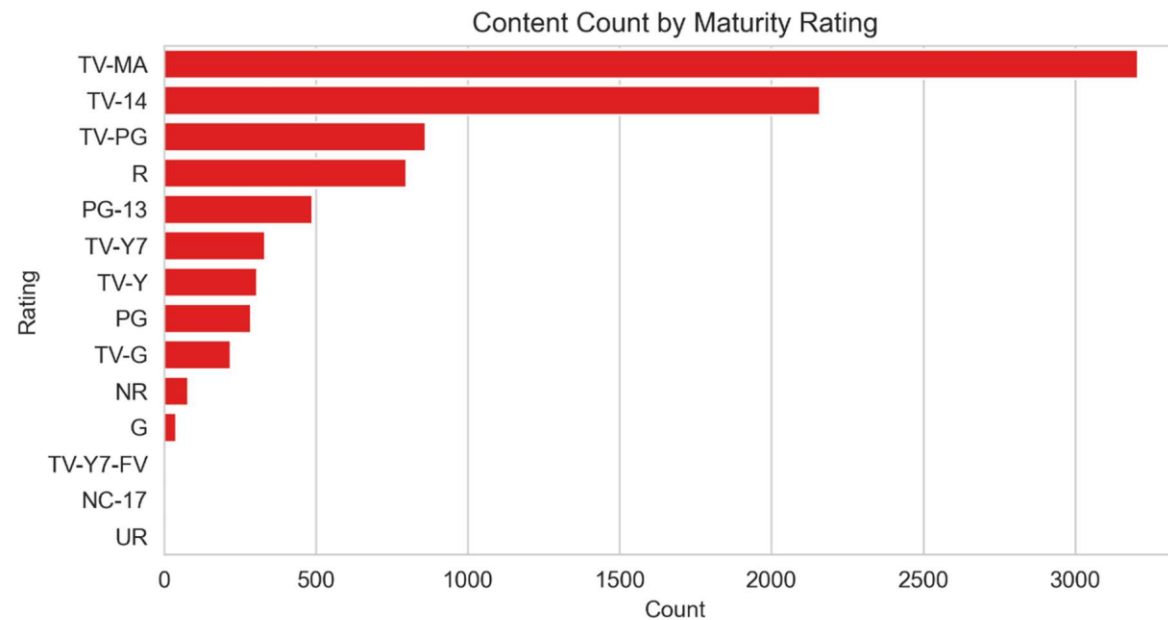


Content Count by Maturity Rating

Counting titles by rating reveals which audience segments Netflix serves most.

RESULT TV-MA and TV-14 lead, together making up about 61% of the catalogue.

APPROACH Counted titles per rating category and plotted them as a horizontal bar chart, sorted descending.

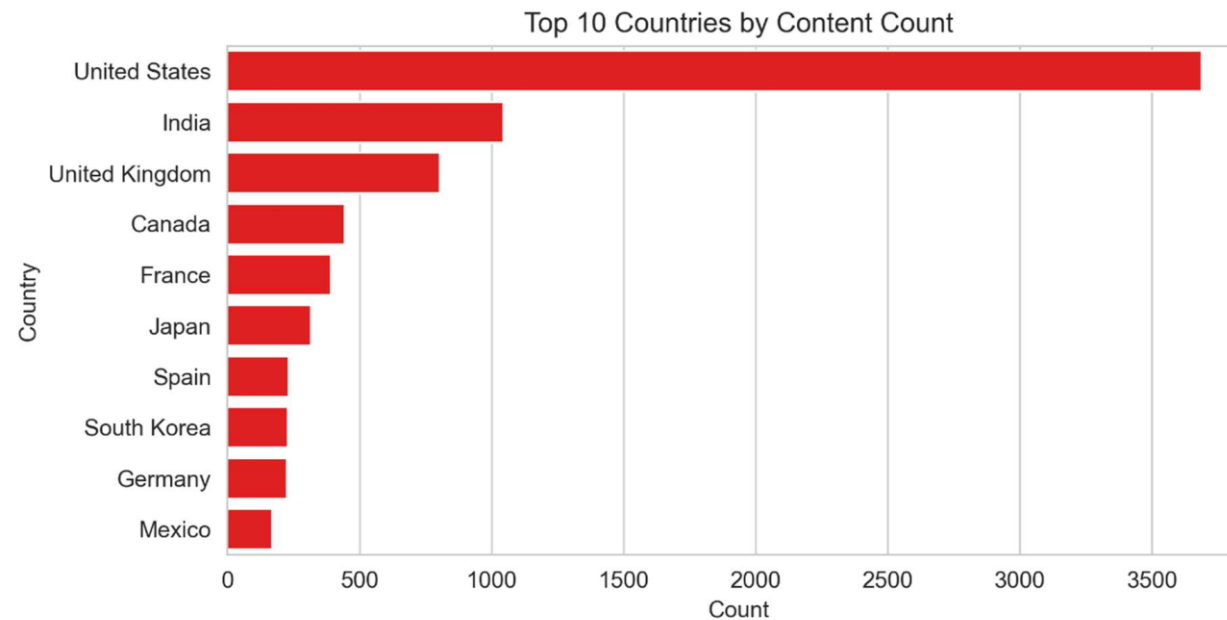


Top 10 Countries by Content Count

The country field was unnested (comma-separated) before counting, so co-productions count once per country.

RESULT The United States leads with 3,689 titles, followed by India (1,046) and the UK (804).

APPROACH Split multi-value country strings into one row per country, counted titles per country, and ranked the top 10.

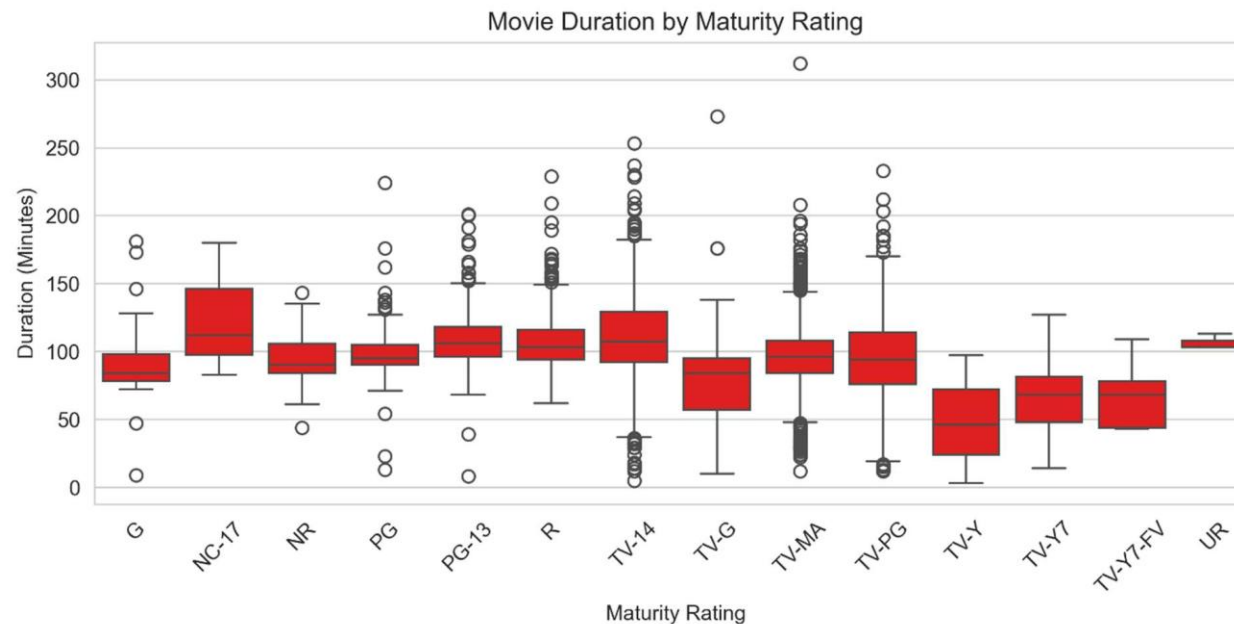


Movie Duration by Maturity Rating

A boxplot compares runtime distributions across rating categories.

RESULT NC-17 and TV-14 movies run longest on average; TV-PG and PG titles are generally shorter.

APPROACH Grouped movie duration by rating and plotted boxplots to compare medians, spread, and outliers per category.

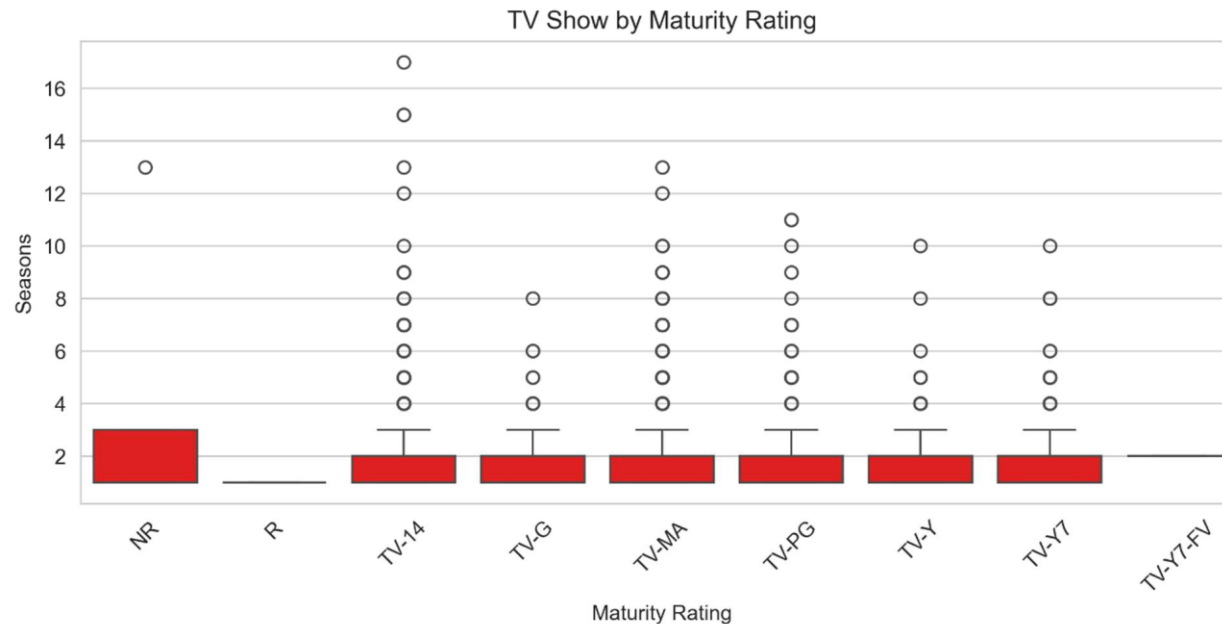


TV Show Seasons by Maturity Rating

The same boxplot approach applied to season count instead of movie runtime.

RESULT Median is 1 season across almost every rating; TV-MA and TV-14 show the widest spread, including 10+ season runs.

APPROACH Grouped season count by rating and plotted boxplots to surface which ratings sustain long-running series.

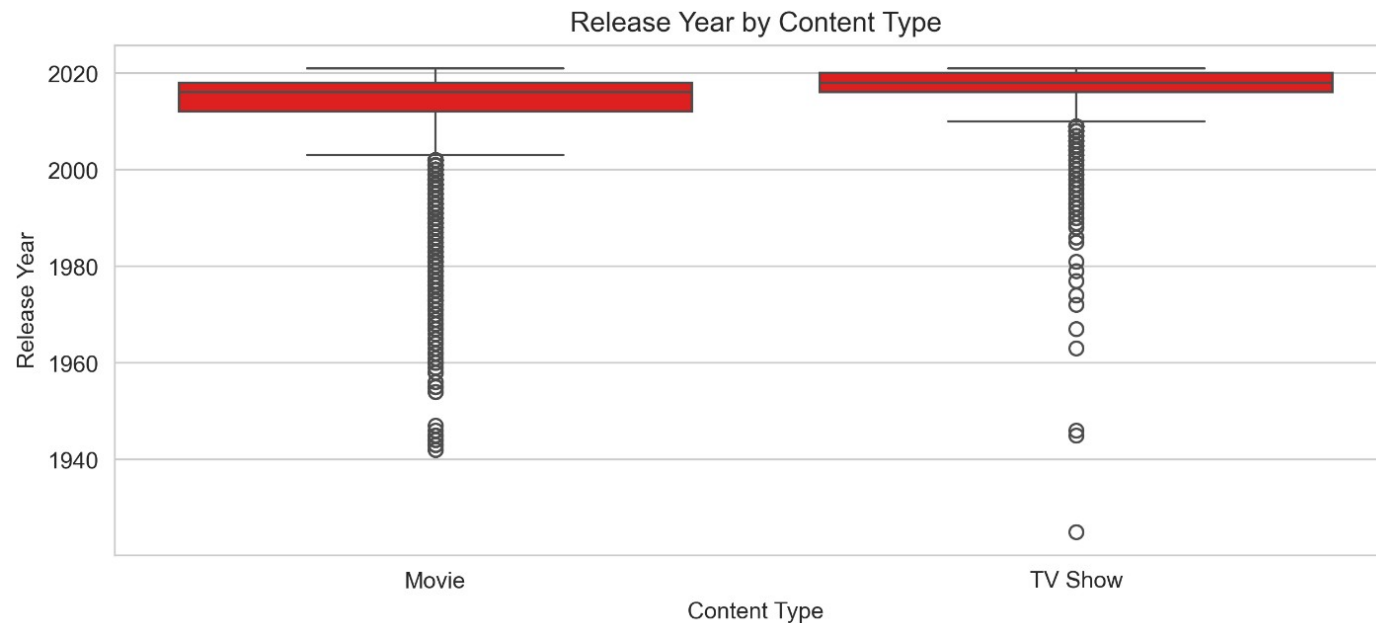


Release Year by Content Type

Comparing `release_year` distributions between Movies and TV Shows shows recency by format.

RESULT TV Shows skew slightly more recent (median 2018) than Movies (median 2016).

APPROACH Split `release_year` by type and plotted side-by-side boxplots to compare medians and spread.

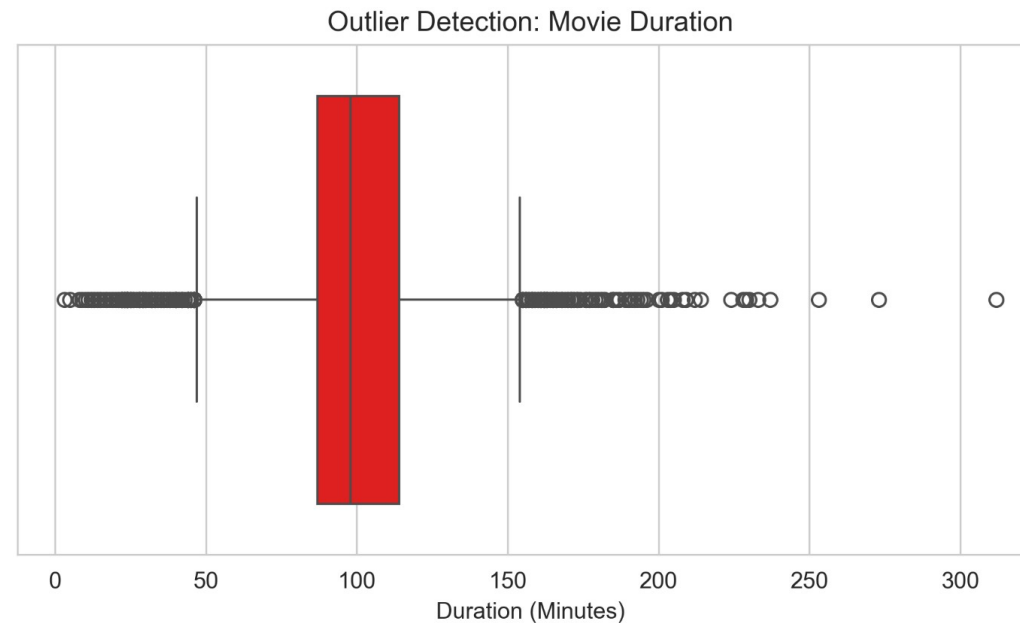


Outlier Detection: Movie Duration

A boxplot on movie runtime flags unusually short or long titles beyond the IQR whiskers.

RESULT A handful of movies exceed 250 minutes — rare specials or epics rather than the norm; these were kept, not removed.

APPROACH Plotted a horizontal boxplot on movie duration and visually inspected points beyond the whiskers as candidate outliers.

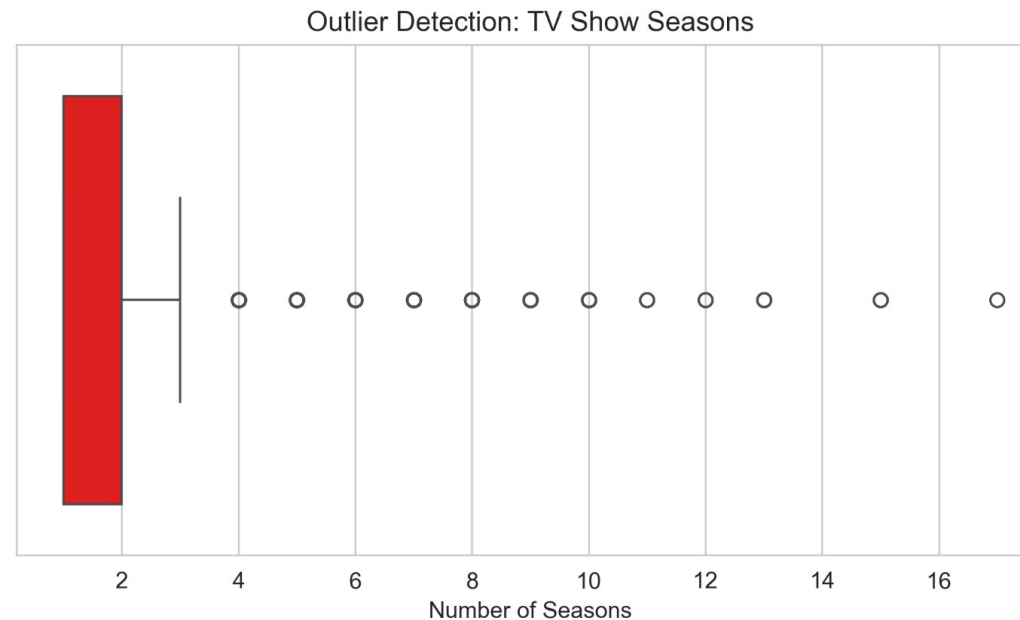


Outlier Detection: TV Show Seasons

The same outlier check applied to season count for TV Shows.

RESULT A small number of long-running series (up to 17 seasons) sit well beyond the whiskers — valid data, not errors.

APPROACH Plotted a boxplot on season count; confirmed high-season outliers were genuine long-running shows and retained them.



Key Business Insights

1

Movies still dominate, but TV Shows are gaining ground.

TV Show additions grew from 25% to 33.7% of new titles between 2018 and 2021, while Movie additions declined after 2020.

2

Long-running series are the exception, not the rule.

About 67% of TV Shows have just one season, and long-running shows (4+ seasons) are concentrated in the TV-MA and TV-14 ratings.

3

Content preferences vary sharply by country.

South Korea and Japan lean toward TV Shows, while India and Germany lean toward Movies — a case for market-specific content strategy.

4

The catalogue is concentrated in a few markets and genres.

The US and India account for a disproportionate share of titles, and Drama, Comedy, and International content dominate the genre mix.

Key Recommendations



Increase investment in TV Shows.

Their share of new additions is growing and they drive longer viewer engagement than one-off movies.



Expand family-friendly content.

The catalogue skews heavily toward TV-MA and TV-14; broadening PG/TV-Y content can widen the audience base.



Tailor content strategy by country.

Grow TV Show investment in South Korea and Japan; continue strong Movie investment in India and Germany.



Prioritize high-quality limited series.

Since most shows run one season, focused limited series can attract viewers at lower long-term production cost.



Time major releases for July and December.

These months see the highest historical content additions and likely audience activity.



Diversify beyond the top markets and genres.

Invest in underrepresented countries and genres to broaden reach and reduce concentration risk.

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THANK YOU.